

Time allowed: 1 hour

All questions may be attempted.

Question 1

a) Find (i) $\int (5x-3)^{15} dx$ 2

(ii) $\int \frac{5}{\sqrt{3x+1}} dx$ 3

b) If $f(x) = \sqrt{\frac{2x-1}{4x+3}}$ then $f'(x) = \frac{5}{\sqrt{(2x-1)(4x+3)^3}}$. (You do not need to prove this!)

Use this fact to evaluate $\int_{3/2}^{1/2} \frac{dx}{\sqrt{(2x-1)(4x+3)^3}}$ 3

Question 2

a) Find the area enclosed by the curve $y = 3 + 2x - x^2$, the x-axis and the lines $x = -1$ and $x = 2$. 3

b) Calculate the area between the curve $y = x^3 - x^2 - 2x$ and the x-axis. 5

c) The area bounded by the curve $y = x^3$, the y-axis and the lines $y = 0$ and $y = k$ is 12 units. Find the value of k . 5

d) Find the area enclosed by the curve $y = \frac{1}{4}x^3$ and the line $y = x$. 4

Question 3

Calculate the volume of the solid generated by rotating the ellipse $\frac{x^2}{4} + y^2 = 1$ about the x-axis. 4

Question 4

Calculate an approximation for the area between the curve $y = 2^x$, the x-axis and the lines $x = 1$, $x = 3$

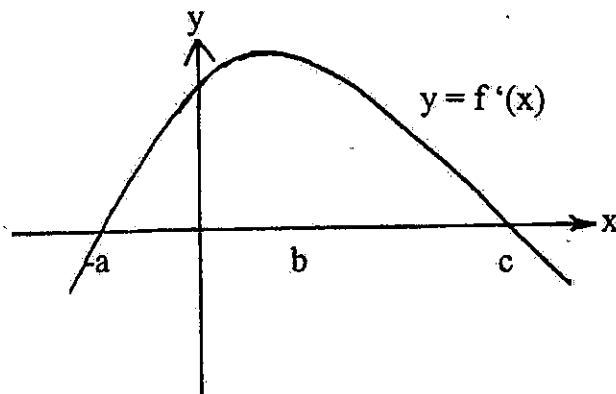
- a) by dividing the area into two strips and using the trapezoidal rule 2
- b) by using Simpson's rule with three function values. 2

Question 5

For the curve $y = \frac{x^2}{x^2 - 4}$

- a) Show that there is a maximum turning point at $(0,0)$. 3
- b) Find, showing working, any
 - (i) vertical asymptotes 2
 - (ii) horizontal asymptotes. 2
- c) Is the function even or odd, give reasons? 1
- d) Draw a neat sketch of the function showing its main features. 2

Question 6



The parabola drawn in the diagram above shows the graph of the gradient function $y = f'(x)$ for a given function $y = f(x)$. On separate diagrams draw possible graphs of

- a) $y = f''(x)$ 2
- b) $y = f(x)$ 3

Question 1

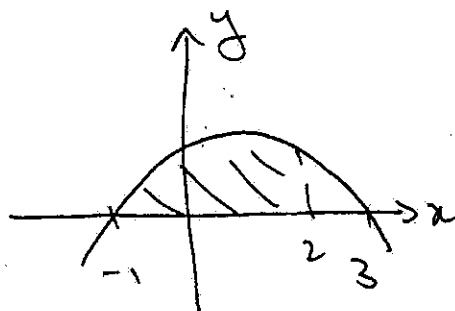
$$\begin{aligned} \text{a) (i)} \int (5x-3)^{15} dx &= \frac{1}{5} \times \frac{(5x-3)^{16}}{16} + C \\ &= \frac{(5x-3)^{16}}{80} + C \end{aligned}$$

$$\begin{aligned} \text{(ii)} \int \frac{5}{\sqrt{3x+1}} dx &= \int 5(3x+1)^{-\frac{1}{2}} dx \\ &= 5 \times \frac{(3x+1)^{\frac{1}{2}}}{\frac{1}{2}} \times \frac{1}{3} + C \\ &= \frac{10 \sqrt{3x+1}}{3} + C \end{aligned}$$

$$\begin{aligned} \text{b) } \int_{1\frac{1}{2}}^{5\frac{1}{2}} \frac{dx}{\sqrt{(2x-1)(4x+3)^3}} &= \frac{1}{5} \left[\sqrt{\frac{2x-1}{4x+3}} \right]_{1\frac{1}{2}}^{5\frac{1}{2}} \\ &= \frac{1}{5} \left[\sqrt{\frac{10}{25}} - \sqrt{\frac{2}{9}} \right] \\ &= \frac{1}{5} \left[\frac{\sqrt{10}}{5} - \frac{\sqrt{2}}{3} \right] \end{aligned}$$

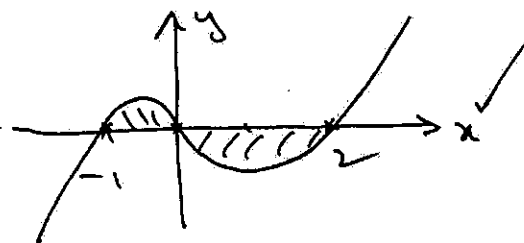
Question 2

$$\begin{aligned} \text{a)} \quad & 3+2x-x^2=0 \\ & x^2-2x-3=0 \\ & (x+1)(x-3)=0 \\ & x=-1, 3 \end{aligned}$$



$$\begin{aligned} A &= \int_{-1}^3 (3+2x-x^2) dx \\ &= \left[3x + x^2 - \frac{x^3}{3} \right]_{-1}^3 \\ &= \left(6 + 4 - \frac{8}{3} \right) - \left(-3 + 1 + \frac{1}{3} \right) \\ &= 9 \end{aligned}$$

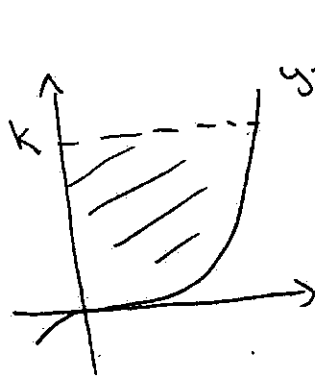
$$\begin{aligned} \text{b)} \quad & x(x^2-x-2)=0 \\ & x(x+1)(x-2)=0 \\ & x=0, -1, 2 \end{aligned}$$



$$\begin{aligned} A &= \int_{-1}^0 (x^3 - x^2 - 2x) dx + \left| \int_0^2 (x^3 - x^2 - 2x) dx \right| \\ &= \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_{-1}^0 + \left| \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_0^2 \right| \\ &= \left[0 - \left(\frac{1}{4} + \frac{1}{3} - 1 \right) \right] + \left| \left[\left(4 - \frac{8}{3} - 4 \right) - 0 \right] \right| \\ &= \frac{5}{12} + \left| \left(-\frac{8}{3} \right) \right| \\ &= \frac{37}{12} \end{aligned}$$

Q.2

c)



$$y = x^3 \Rightarrow x = y^{\frac{1}{3}}$$

$$\int_0^k y^{\frac{1}{3}} dy = 12 \checkmark$$

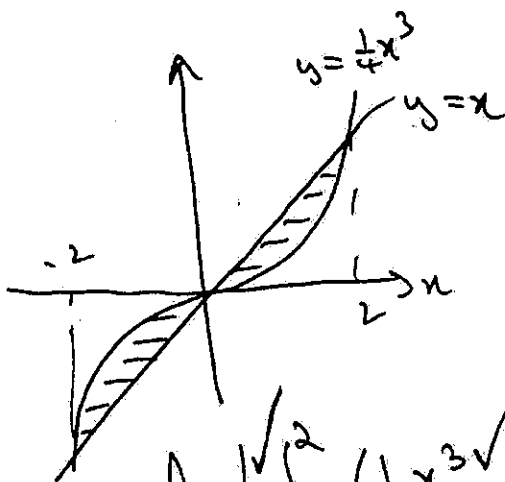
$$\left[\frac{3}{4} y^{\frac{4}{3}} \right]_0^k = 12$$

$$\frac{3}{4} k^{\frac{4}{3}} = 12$$

$$k^{\frac{4}{3}} = 16 \checkmark$$

$$k = 8 \checkmark$$

d)



$$\frac{1}{4} x^3 = x$$

$$\frac{1}{4} x^3 - x = 0$$

$$x \left(\frac{1}{4} x^2 - 1 \right) = 0$$

$$x = 0, x = \pm 2 \checkmark$$

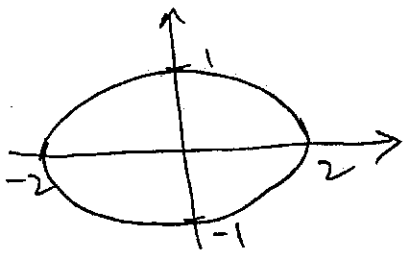
$$A = \left| \frac{1}{2} \int_{-2}^2 \left(\frac{1}{4} x^3 - x \right) dx \right|$$

$$= \left| \frac{1}{2} \left[\frac{x^4}{16} - \frac{x^2}{2} \right]_{-2}^2 \right|$$

$$= \left| \frac{1}{2} [1 - 2] \right|$$

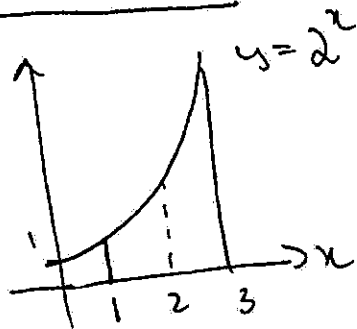
$$= 2 \checkmark$$

Question 3



$$\begin{aligned}
 V &= \pi \int_{-2}^2 \left(1 - \frac{x^2}{4}\right) dx \quad \text{limits} \\
 &= 2\pi \int_0^2 \left(1 - \frac{x^2}{4}\right) dx \\
 &= 2\pi \left[x - \frac{x^3}{12} \right]_0^2 \\
 &= 2\pi \left[\left(2 - \frac{8}{12}\right) - 0 \right] \\
 &= 8\pi \frac{2}{3}
 \end{aligned}$$

Question 4



$$\begin{aligned}
 \text{Area trapez. } \checkmark \\
 a) A &\doteq \frac{1}{2} (2^1 + 2^2) + \frac{1}{2} (2^2 + 2^3) \\
 &= 9 \text{ u}^2 \checkmark
 \end{aligned}$$

$$\begin{aligned}
 b) A &= \frac{3-1}{6} \{2^1 + 4 \times 2^2 + 2^3\} \checkmark \\
 &= \frac{1}{3} \{26\} \\
 &= 8\frac{2}{3} \text{ u}^2 \checkmark
 \end{aligned}$$

Question 5

$$a) \quad y' = \frac{(x^2-4) \cdot 2x - x^2 \cdot 2x}{(x^2-4)^2}$$

$$= \frac{2x^3 - 8x - 2x^3}{(x^2-4)^2}$$

$$= \frac{-8x}{(x^2-4)^2} \quad \checkmark$$

$$y' = 0 \Rightarrow x = 0 \quad \checkmark$$

x	-1	0	1	
y'	$+\frac{8}{9}$	0	$-\frac{8}{9}$	$\checkmark$

$\therefore$ Max t. p. at $(0,0)$

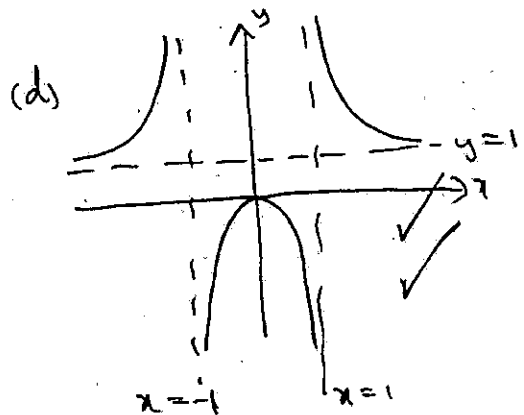
b) (i) Vert. asymptotes
where $x^2 - 4 = 0$ $\checkmark$
ie. $x = \pm 2$ $\checkmark$

$$(ii) \quad \lim_{x \rightarrow \pm \infty} \frac{x^2}{x^2-4} \checkmark = \lim_{x \rightarrow \pm \infty} \frac{1}{1 - \frac{4}{x^2}} \\ = 1 \quad \checkmark$$

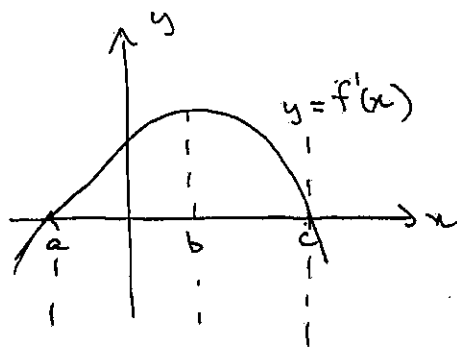
ie. $y = 1$ is horizontal asymptote

$$c) \quad f(-x) = \frac{(-x)^2}{(-x)^2 - 4} \\ = \frac{x^2}{x^2 - 4} = f(x) \quad \checkmark$$

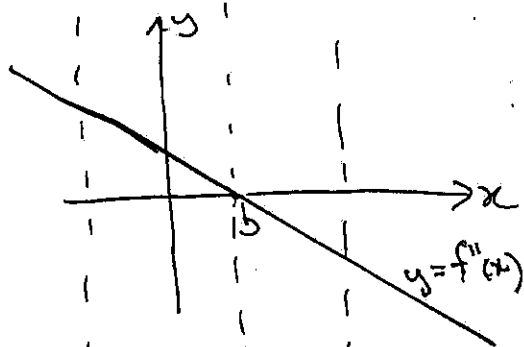
$\therefore y = f(x)$ is an even fn.



Q.6

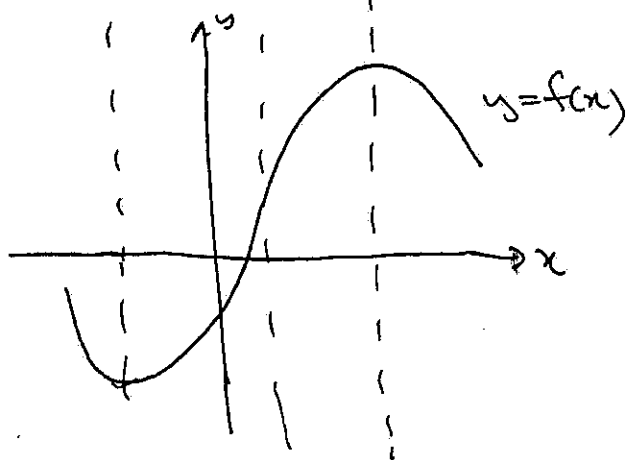


a)



St. Line correct slope ✓
Through $(b, 0)$ ✓

b)



Turning pts at $x = a$ (Min) ✓
 $x = b$ (Max) ✓

Inflection at $x = b$ ✓